

Pregnant Iron Overdose



Section I: Scenario Demographics

Scenario Title:	Iron Overdose in Pregnant Patient		
Date of Development:	10/04/2017		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. Dawn Lim / Dr. Kate Hayman
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To allow learners to investigate and treat a severe, rare overdose in an undifferentiated, unstable patient
CRM Objectives:	1. Manage resources to complete simultaneous investigation and treatment of an unstable patient 2. Maintain situational awareness while integrating information from several different places
Medical Objectives:	1. Stabilize a hemodynamically unstable pregnant patient 2. Work through the broad differential for undifferentiated shock 3. Initiate chelation therapy in iron toxicity

Case Summary: Brief Summary of Case Progression and Major Events

A 29-year old woman with a history of depression and an early unplanned pregnancy is found at home with decreased level of consciousness. She comes to the ED with EMS and her boyfriend. She remains altered in the resuscitation room and declines despite aggressive resuscitation.

After gathering history from the boyfriend, it seems likely that she has ingested a large quantity of pre-natal vitamins resulting in iron toxicity. This is confirmed on bloodwork and imaging. She will require airway management, hemodynamic support and specific chelation therapy.

References

- Marx JA, Hockberger RS, Walls RM, & Adams J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Hoffman RS, Howland MA, Lewin NA, Nelson LS & Goldfrank LR. (2015). *Goldfrank's Toxicologic Emergencies*. New York: McGraw-Hill.
- Nickson, C. (2014). *Iron Overdose*. Life in the Fast Lane. Retrieved from: <https://lifeinthefastlane.com/ccc/iron-overdose/>

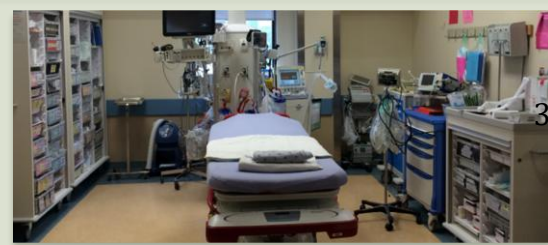
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Boyfriend	Worried but helpful. Does not know what happened overnight. Found the patient lethargic and confused this morning. Brought in patient's pill bottles. Only gives information when directly asked.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Female patient sitting-up, gown on with pants, monitors not attached, IV x1 in place.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in a large community ED. You are called to a resuscitation room where EMS has just brought in a 29-year woman with altered mental status. Her boyfriend called 9-1-1 when he found her confused this morning. She is 10 weeks pregnant and had some vomiting and diarrhea yesterday. Her boyfriend is in the waiting room.

B. Patient Profile and History

Patient Name: Shelley Age: 29 Weight: 60

Gender: ☐ M ☒ F Code Status: Full

Chief Complaint: Confusion/Vomiting/Diarrhea

History of Presenting Illness: **Pt unable to give history.**

From boyfriend: Pt is 10 weeks pregnant. She started having vomiting and diarrhea yesterday evening. She thinks she saw some streaks of blood in her diarrhea.

Explains that they had an argument yesterday because they have been stressed over this unplanned and unwanted pregnancy. They don't live together. He left after lunch.

She called at night to tell him she had vomiting and diarrhea, but told him not to come over because she was still upset. He came over to see how she was this morning. When he found her in bed, lethargic and confused, he called EMS.

She did not complain of dysuria, frequency, fever. She noticed no vaginal bleeding.

She had used alcohol and marijuana in the past but none in the last 4 weeks since she learned she was pregnant. Has been taking Prozac and a prenatal vitamin daily.

Past Medical History:	Anxiety/depression	Medications:	Fluoxetine
	Irritable bowel syndrome		Pre-natal vitamins
	1 st trimester pregnancy		

Allergies: PCN

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display ☒ Monitor On, no data displayed ☐ Monitor on Standard Display

HR: 130/min BP: 95/60 RR: 30/min O₂SAT: 94% RA

Rhythm: Sinus tach T: 37°C Glucose: 17.5 mmol/L GCS: 9 (E2 V2 M5)

General Status: Toxic and unwell

CNS: Lethargic and moaning. PERL 3mm.

CVS: Tachycardic, heart sounds normal.

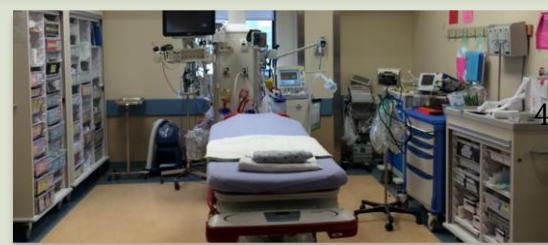
RESP: Tachypneic, breath sounds normal.

ABDO: Diffuse tenderness, no rigidity or guarding.

MSK: Nil. SKIN: Scattered bruises



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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tachycardia HR: 130/min BP: 90/60 RR: 28/min O ₂ SAT: 94% RA T: 37°C	Drowsy. Moaning. Laboured breathing.	<u>Learner Actions</u> ☐ History & physical ☐ Monitors ☐ Ask for med list ☐ Apply O ₂ ☐ 2 nd PIV ☐ 1-2L saline bolus ☐ Tox/Abdo/G+S Bloodwork ☐ B-HCG ☐ Cap glucose ☐ POCUS (IUP, no free fluid) ☐ CXR/AXR	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> O ₂ applied → O ₂ SAT 96% <u>Triggers</u> <i>For progression to next state</i> - All actions complete or 5 min into case → 2. Worsening hemodynamics
2. Worsening hemodynamics HR → 140 BP → 80/50 RR → 25 O ₂ SAT → 88%	Breathing more labored. Becomes obtunded. **VBG result given at onset of state	<u>Learner Actions</u> ☐ Prepare for intubation ☐ Fluid resuscitation ☐ Push dose pressors ☐ ± Start or prepare vasopressor infusion ☐ Consider naloxone	<u>Modifiers</u> - If not yet asked, boyfriend comes to room - If no intubation by 8 min , O ₂ SAT drops → 80% <u>Triggers</u> - Intubate → 3. Persistent hypotension
3. Persistent hypotension HR → 110 BP → 80/55 RR → 12 O ₂ SAT → 96%	Easy intubation. Sedated.	<u>Learner Actions</u> ☐ Call poison control ☐ Start vasopressor or increase dose ☐ Start chelation therapy ☐ Prepare for whole bowel irrigation (insert OG) ☐ Foley catheter ☐ Post-intubation sedation ☐ Heavy metal levels	<u>Modifiers</u> - If not yet asked, boyfriend shows bag of meds (1 empty bottle of prenatal vitamins, full Prozac and herbal supplements) - Rest of bloodwork given - If called, Poison Control suggests deferoxamine at 15mg/kg/hr (may lower for hypotension) and whole bowel irrigation <u>Triggers</u> - Start chelation → 4. Resolution
4. Resolution HR → 95 BP → 105/65 RR → 12 O ₂ SAT → 96%	Sedated. Intubated. Ventilated.	<u>Learner Actions</u> ☐ Call ICU ☐ Discuss psych assessment	<u>Modifiers</u> <u>Triggers</u> - ICU called → End of case

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

VBG: pH 7.05/PCO₂ 38/PO₂ 80/HCO₃ 5

CBC: Hb 90/WBC 17/Plts 65

Lytes: Na 135/K + 4/HCO₃ 5/Cl 100/ Cr 130/Glucose 17

Lactate: 8

LFTs: ALT 560/AST 512/Alk phos 300/Bili 43/ INR 2.3

Tox: APAP <10/ASA neg/EtOH neg

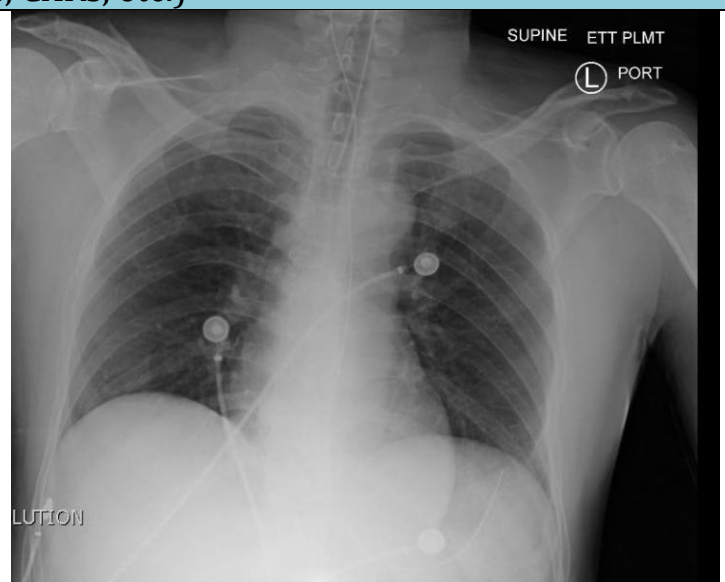
Iron level: 70 micromol/L

Heavy metal/toxic alcohols: Pending

Images (ECGs, CXRs, etc.)



<https://lifeinthefastlane.com/top-ten-foreign-bodies/>



<http://jetem.org/ettcxr/>

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EACH TABLET CONTAINS:	CHAQUE COMPRIMÉ CONTIENT :
VITAMINS	VITAMINES
Beta-Carotene (a source of vitamin A).....1500 mcg/µg / 2500 IU/UI.....	Bêta-carotène (une source de vitamine A).....
Vitamin A (vitamin A acetate) ...300 mcg/µg / 1000 IU/UI.....	Vitamine A (acétate de vitamine A).....
Vitamin E (dl-α tocopheryl acetate).....13.5 mg / 30 IU/UI.....	Vitamine E (acétate de dl-α tocophérol).....
Vitamin C (ascorbic acid).....85 mg.....	Vitamine C (acide ascorbique).....
Folic Acid (folate).....1 mg.....	Acide folique (folate).....
Vitamin B1 (thiamine mononitrate).....1.4 mg.....	Vitamine B1 (mononitrate de thiamine).....
Vitamin B2 (riboflavin).....1.4 mg.....	Vitamine B2 (riboflavine).....
Niacinamide.....18 mg.....	Niacinamide.....
Vitamin B6 (pyridoxine hydrochloride).....1.9 mg.....	Vitamine B6 (chlorhydrate de pyridoxine).....
Vitamin B12 (cyanocobalamin).....2.6 mcg/µg.....	Vitamine B12 (cyanocobalamine).....
Vitamin D (cholecalciferol)10 mcg/µg / 400 IU/UI.....	Vitamine D (cholécalfcérol).....
Biotin.....30 mcg/µg.....	Biotine.....
Pantothenic Acid (calcium d-pantothenate).....6 mg.....	Acide pantothénique (d-pantothénate de calcium).....
MINERALS	MINÉRAUX
Calcium (calcium carbonate).....250 mg.....	Calcium (carbonate de calcium).....
Magnesium (magnesium oxide).....50 mg.....	Magnésium (oxyde de magnésium).....
Iodine (potassium iodide).....220 mcg/µg.....	Iode (iodure de potassium).....
Iron (ferrous fumarate).....27 mg.....	Fer (fumarate ferreux).....
Copper (cupric sulphate).....1 mg.....	Cuivre (sulfate de cuivre).....
Chromium (chromium chloride).....30 mcg/µg.....	Chrome (chlorure chromique).....
Manganese (manganese sulphate).....2 mg.....	Manganèse (sulfate de manganèse).....
Molybdenum (sodium molybdate).....50 mcg/µg.....	Molybdène (molybdate de sodium).....
Selenium (sodium selenate).....30 mcg/µg.....	Sélénium (séléniate de sodium).....
Zinc (zinc oxide).....7.5 mg.....	Zinc (oxyde de zinc).....
When taken daily prior to becoming pregnant and during early pregnancy	* Lorsque pris tous les jours avant de tomber enceinte et durant les premiers stades de la grossesse

Total possible dose of iron = 27 mg of elemental iron/pill X 100 tabs/55 kg = 49 mg/kg

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Ultrasound Video Files (if applicable)



IUP present

<https://radiologykey.com/first-trimester-pregnancy/>



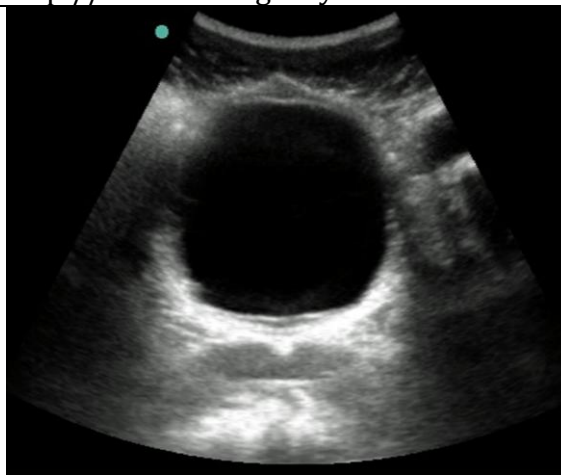
RUQ – no free fluid

<http://www.emergencyultrasoundteaching.com>



RUQ – no free fluid

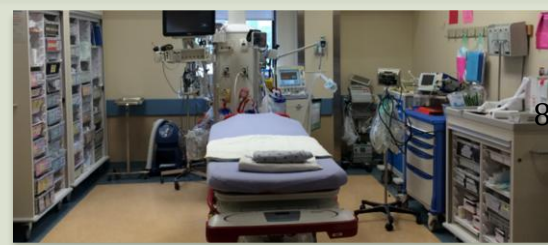
<http://www.emergencyultrasoundteaching.com>



Pelvic – no free fluid

<http://www.emergencyultrasoundteaching.com>

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Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video
Objectives			
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CRM Objectives:	1. Manage resources to complete simultaneous investigation and treatment of an unstable patient 2. Maintain situational awareness while integrating information from several different places		
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Sample Questions for Debriefing			
CRM 1. How can you use the team to effectively gather information while managing an unstable patient? 2. How do you ensure you have gathered all the information you want when working through a broad differential?			
Medical Expert 1. What was your differential diagnosis on the patient's initial presentation? How was this changed by the pregnancy? 2. What are the phases of iron toxicity? 3. How do you estimate the severity of iron toxicity? 4. What are the indications for chelation therapy?			
Key Moments			
Starting with a broad differential diagnosis.			
Recognizing shock and treating aggressively including fluids and intubation.			
Gathering history from the boyfriend to discover the overdose.			
Initiating management of iron toxicity including decontamination and enhanced elimination.			